

CompTIA Linux+ (XK0-005)

Linux Backup and File Compression Concepts

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Introduction

XK0-005

Backup Files

Compress Files

Tar

Dd

Welcome to the **Linux Backup and File Compression Concepts** Practice Lab. In this module, you will be provided with the instructions and devices needed to develop your hands-on skills.

Learning Outcomes

In this module, you will complete the following exercises:

- Exercise 1 - Use Tar to Backup and Compress Files
- Exercise 2 - Use the Dd Command to Backup an Entire Disk

After completing this module, you should be able to:

- Use the Tar Command to Archive and Compress Files
- Backup an Entire Disk using the Dd Command

Exam Objectives

The following exam objectives are covered in this module:

1.2 Given a scenario, manage files and directories

- File compression, archiving, and backup

Lab Duration

It will take approximately **1 hour** to complete this lab.

Help and Support

For more information on using Practice Labs, please see our **Help and Support** page. You can also raise a technical support ticket from this page.

Click **Next** to view the Lab topology used in this module.

Lab Topology

During your session, you will have access to the following lab configuration.

Depending on the exercises, you may or may not use all of the devices, but they are shown here in the layout to get an overall understanding of the topology of the lab.

- **PLABSRV01** - Ubuntu 20.04 - Stand-Alone Ubuntu Server (Console)
- **PLABSRV02** - Alma Linux 8.5 - Stand-Alone Linux Server (GUI)
- **PLABCLIENT01** - Ubuntu 20.04 - Stand-Alone Ubuntu Workstation (GUI)
- **PLABWIN10** - Windows 10 - Stand-Alone Workstation

Click **Next** to proceed to the first exercise.

Exercise 1 - Use Tar to Backup and Compress Files

The tar (Tape Archive) command is used in Linux to create, compress and extract archive files. The archived files can be used for the following purposes:

- The backup of the server and user data

- The archiving of documents for preservation
- Distribution of software in a compressed format

In this exercise, you will use the tar command to archive and compress files.

Learning Outcomes

After completing this exercise, you should be able to:

- Use the Tar Command to Archive and Compress Files

Your Devices

You will be using the following device in this lab. Please power this on now.

- **PLABCLIENT01** - Ubuntu 20.04 - Stand-Alone Ubuntu Workstation (GUI)

Task 1 - Use the Tar Command to Archive and Compress Files

In this task, you will use the tar command to archive multiple files into a single file.

Step 1

Connect to **PLABCLIENT01**.

Login as **administrator** using the following **Password**:

Passw0rd

Press **Enter**.

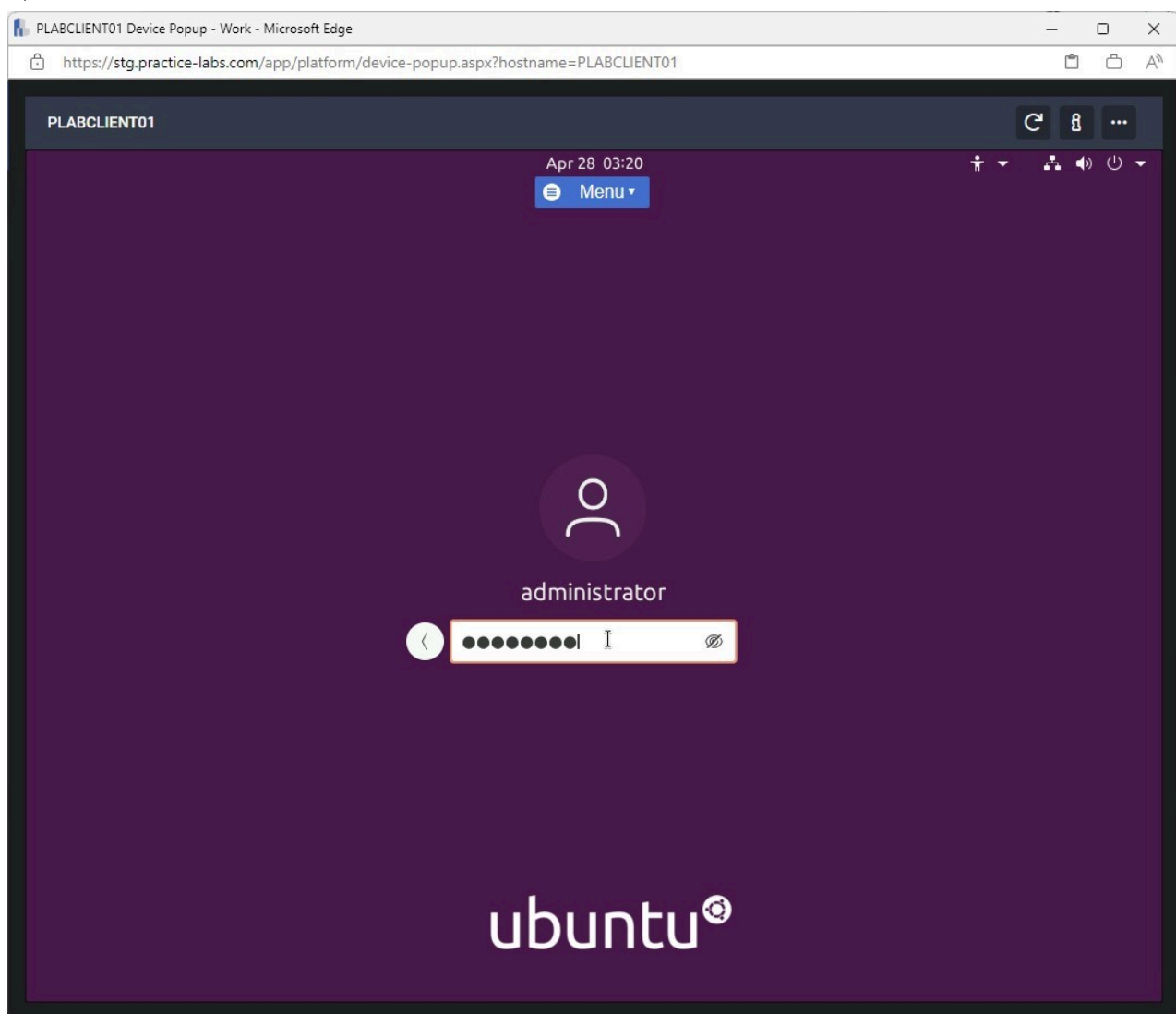


Figure 1.1 Screenshot of PLABCLIENT01: The login screen is displayed, showing the required credentials typed in.

Step 2

Click the **Show applications** icon on the bottom left corner.

Scroll down and select **Terminal**.

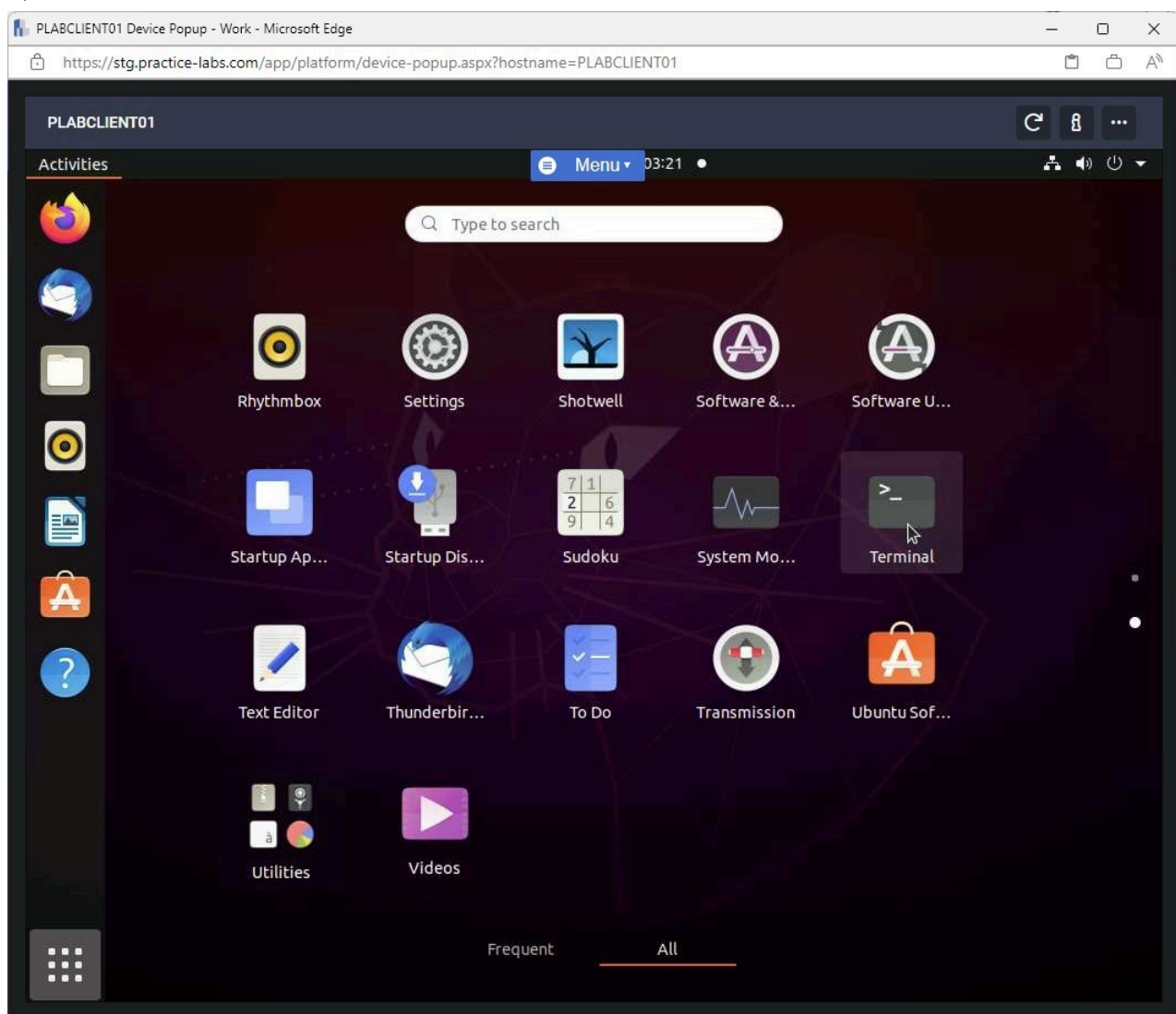


Figure 1.2 Screenshot of PLABCLIENT01: Show applications icon is clicked, and Terminal is selected.

Step 3

In the **Terminal** window, type the following command to navigate to the **Books** directory:

```
cd Books
```

Press **Enter**.

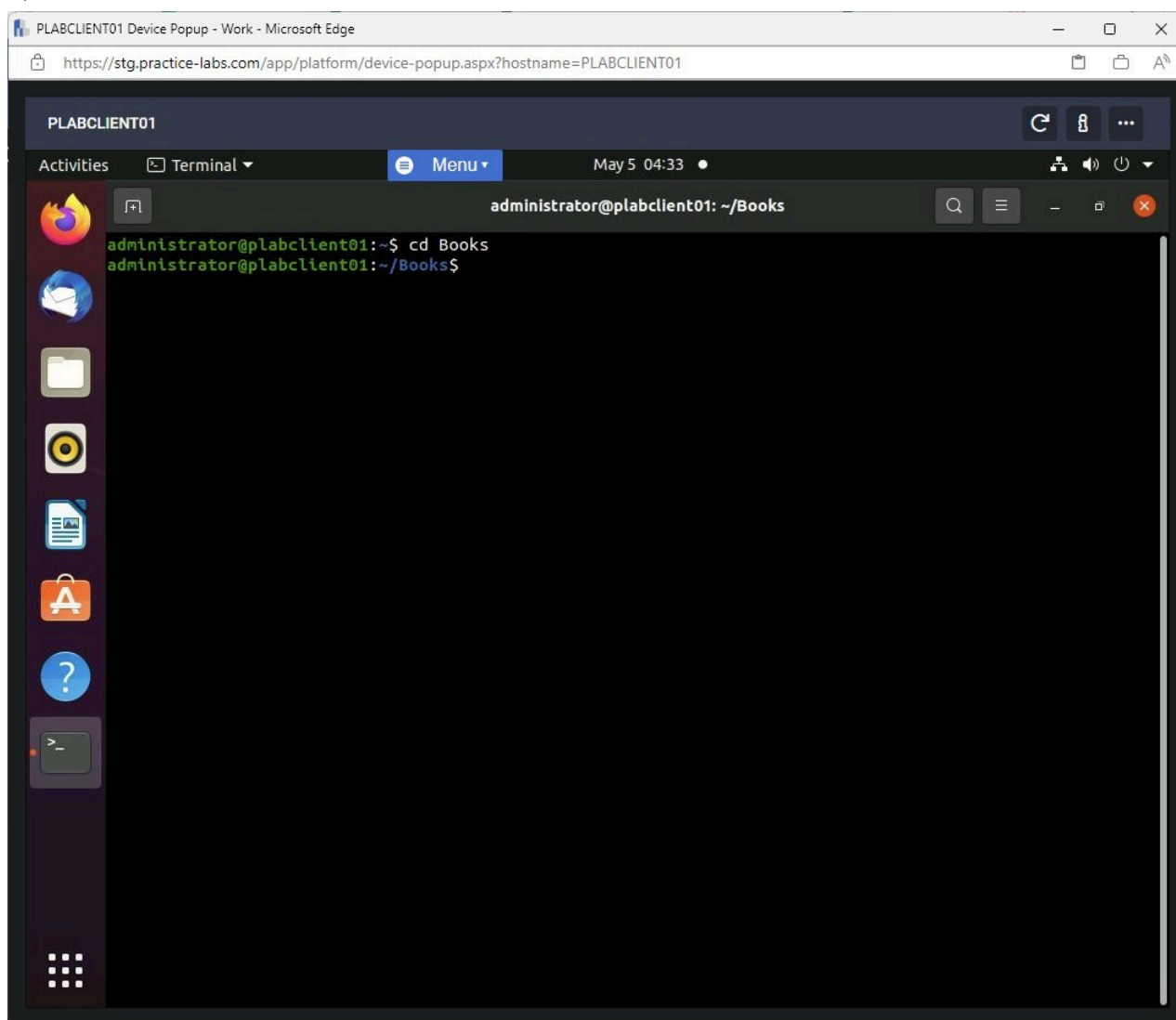


Figure 1.3 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Step 4

In the **Terminal** window, type the following command and press **Enter**:

```
ls -l
```

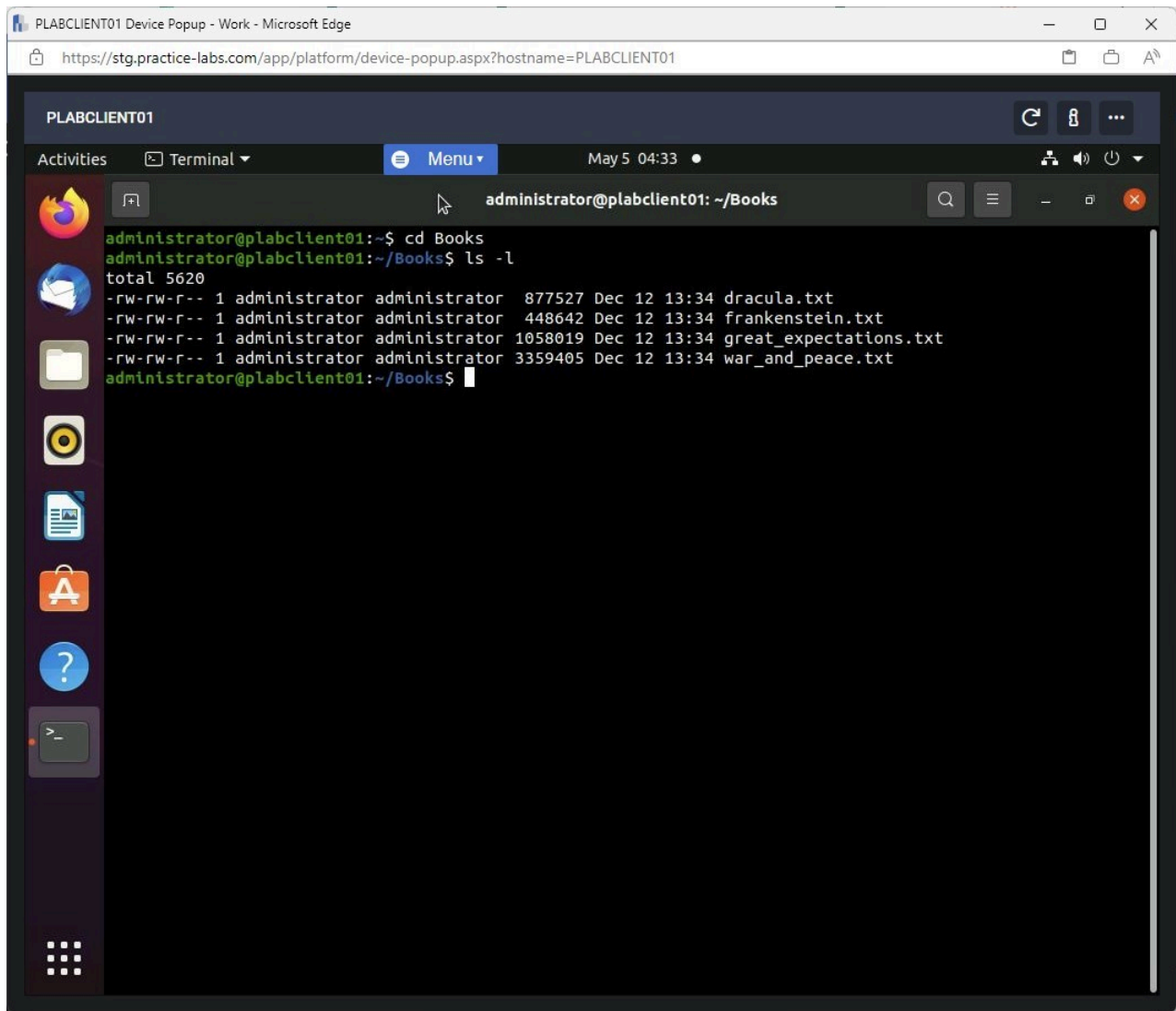


Figure 1.4 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: Notices the file sizes. The file sizes are displayed in bytes units. If you add them up, it would be around 5.7MB.

Step 5

In the **Terminal** window, type the following command to navigate to the Home directory:

```
cd ~
```

Press **Enter**.

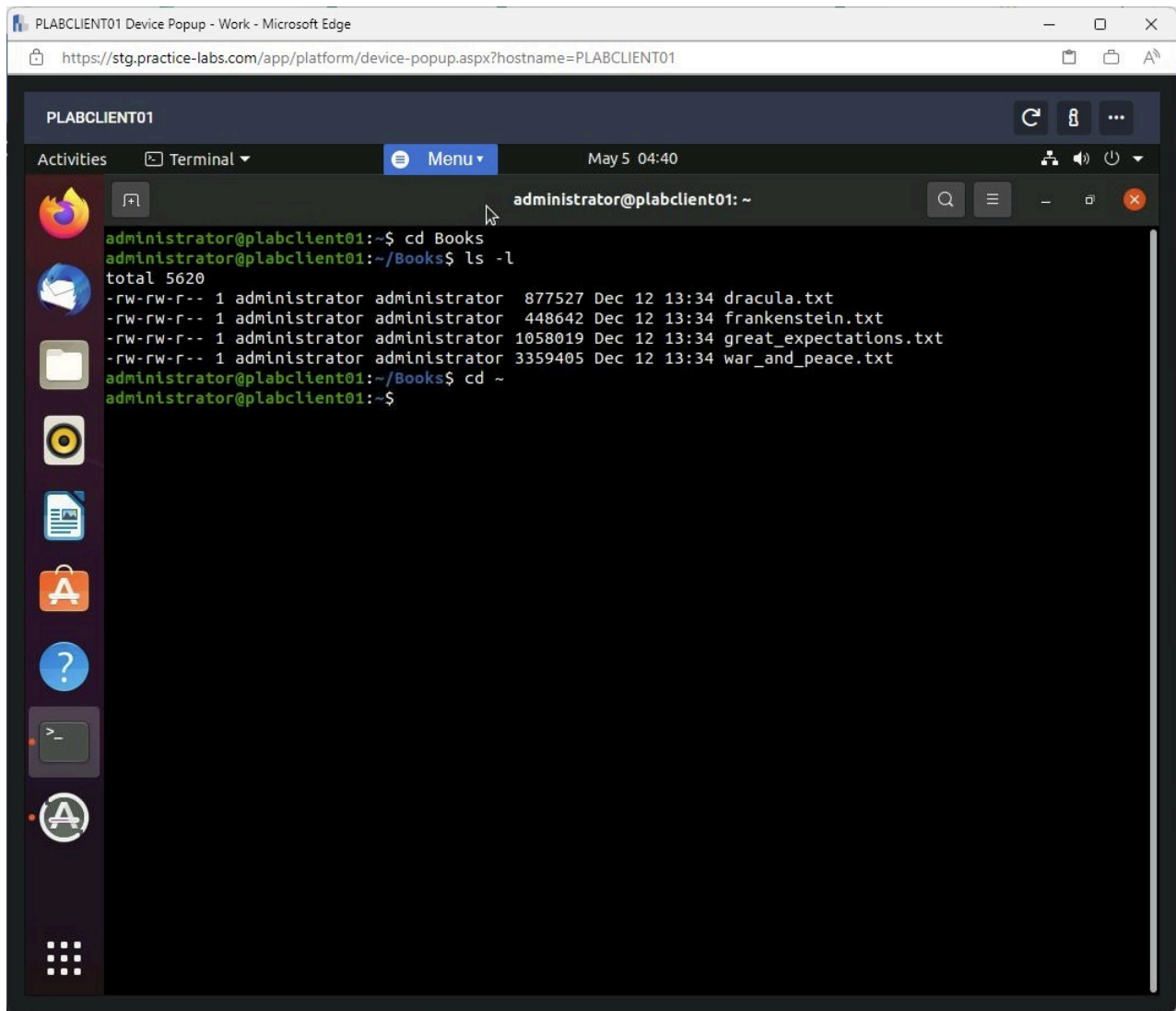


Figure 1.5 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Step 6

In the **Terminal** window, type the following command and press **Enter**:

```
tar -cf book_backup.tar Books
```

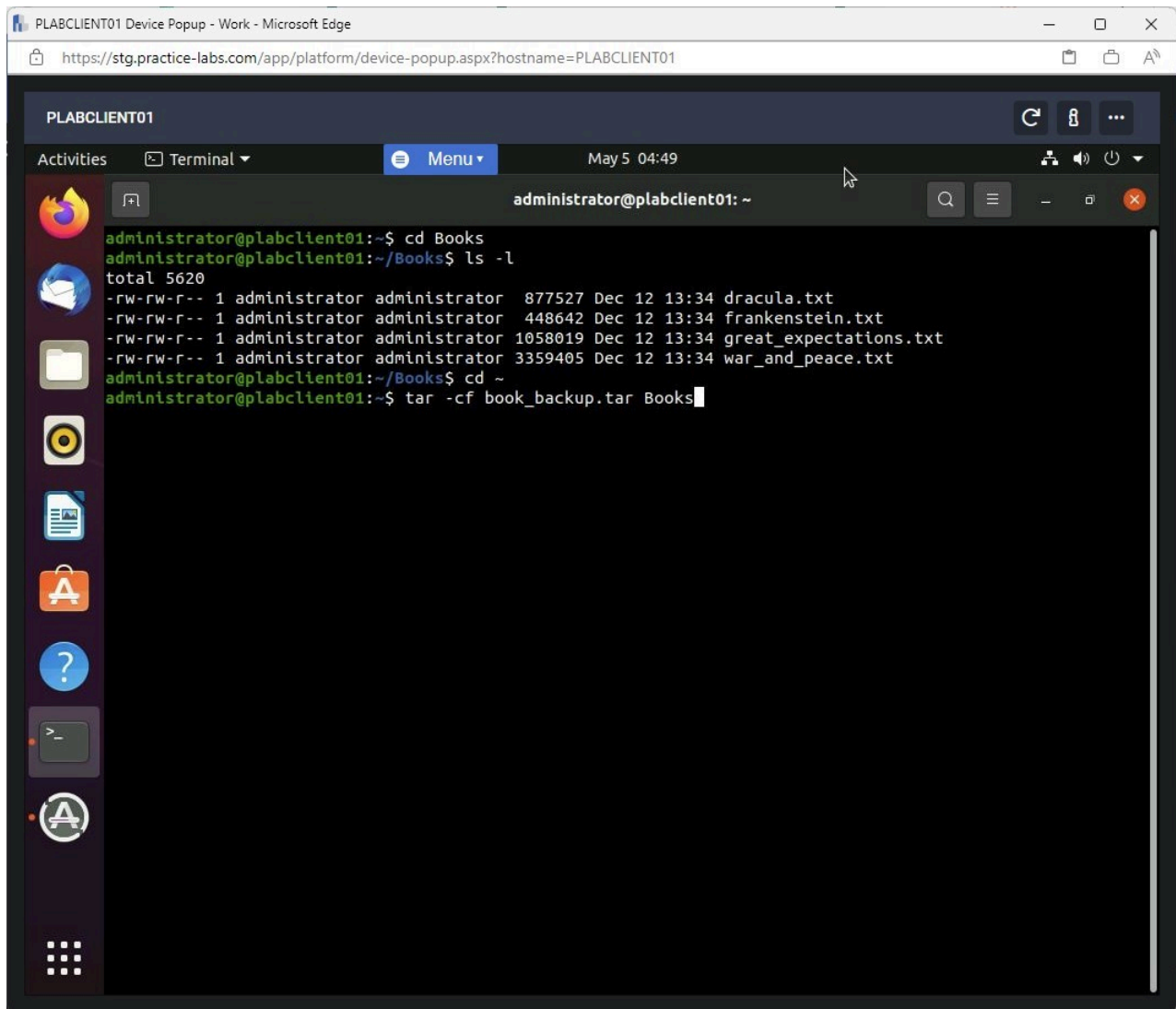


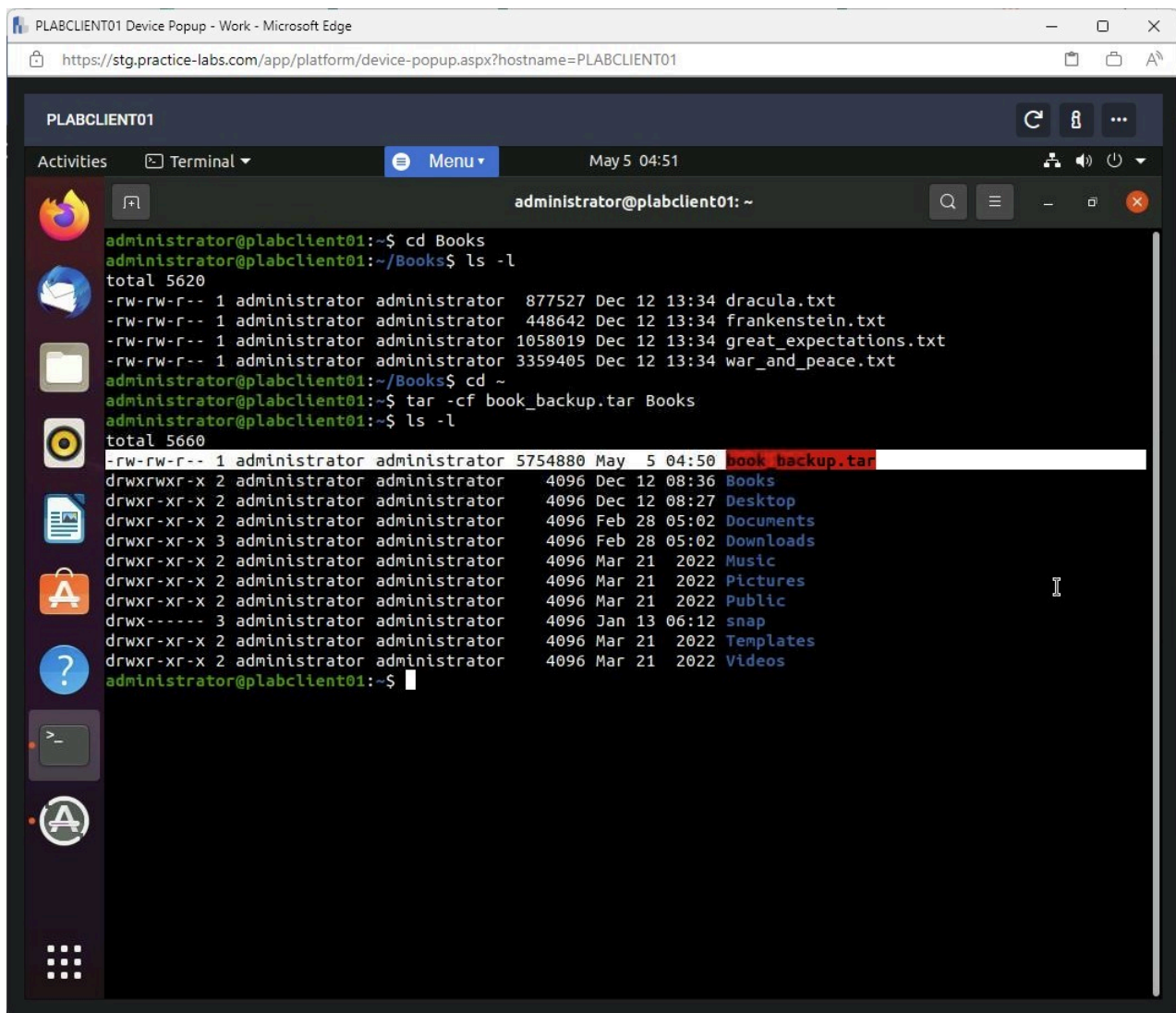
Figure 1.6 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered.

Note: The **tar** command is used to create archive files. The **-c** parameter creates an archive file, and **f** is used to specify the filename for the archive file. In this case, **book_backup.tar**.

Step 7

In the **Terminal** window, type the following command and press **Enter**:

```
ls -l
```



The screenshot shows a terminal window titled 'PLABCLIENT01 Device Popup - Work - Microsoft Edge' with the URL 'https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABCLIENT01'. The terminal interface includes a top bar with 'PLABCLIENT01', a search icon, and a menu icon. Below this is a navigation bar with 'Activities', 'Terminal', and 'Menu' tabs, along with the date 'May 5 04:51'. The terminal window itself has a title bar 'administrator@plabclient01: ~' and a search icon. The command history shows the user navigating to the 'Books' directory and listing its contents. The files listed are 'dracula.txt', 'frankenstein.txt', 'great_expectations.txt', and 'war_and_peace.txt'. The user then runs the command 'tar -cf book_backup.tar Books' to create an archive. The final 'ls -l' command shows the new file 'book_backup.tar' with a size of 5754880 bytes, while the 'Books' directory remains at 4096 bytes.

```
administrator@plabclient01:~$ cd Books
administrator@plabclient01:~/Books$ ls -l
total 5620
-rw-rw-r-- 1 administrator administrator 877527 Dec 12 13:34 dracula.txt
-rw-rw-r-- 1 administrator administrator 448642 Dec 12 13:34 frankenstein.txt
-rw-rw-r-- 1 administrator administrator 1058019 Dec 12 13:34 great_expectations.txt
-rw-rw-r-- 1 administrator administrator 3359405 Dec 12 13:34 war_and_peace.txt
administrator@plabclient01:~/Books$ cd ~
administrator@plabclient01:~$ tar -cf book_backup.tar Books
administrator@plabclient01:~$ ls -l
total 5660
-rw-rw-r-- 1 administrator administrator 5754880 May 5 04:50 book_backup.tar
drwxrwxr-x 2 administrator administrator 4096 Dec 12 08:36 Books
drwxr-xr-x 2 administrator administrator 4096 Dec 12 08:27 Desktop
drwxr-xr-x 2 administrator administrator 4096 Feb 28 05:02 Documents
drwxr-xr-x 3 administrator administrator 4096 Feb 28 05:02 Downloads
drwxr-xr-x 2 administrator administrator 4096 Mar 21 2022 Music
drwxr-xr-x 2 administrator administrator 4096 Mar 21 2022 Pictures
drwxr-xr-x 2 administrator administrator 4096 Mar 21 2022 Public
drwx----- 3 administrator administrator 4096 Jan 13 06:12 snap
drwxr-xr-x 2 administrator administrator 4096 Mar 21 2022 Templates
drwxr-xr-x 2 administrator administrator 4096 Mar 21 2022 Videos
administrator@plabclient01:~$
```

Figure 1.7 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered.

Note: The **book_backup.tar** archive file is now created. Notice the file size has not changed, even though an archive was created. The tar command does not compress files by default.

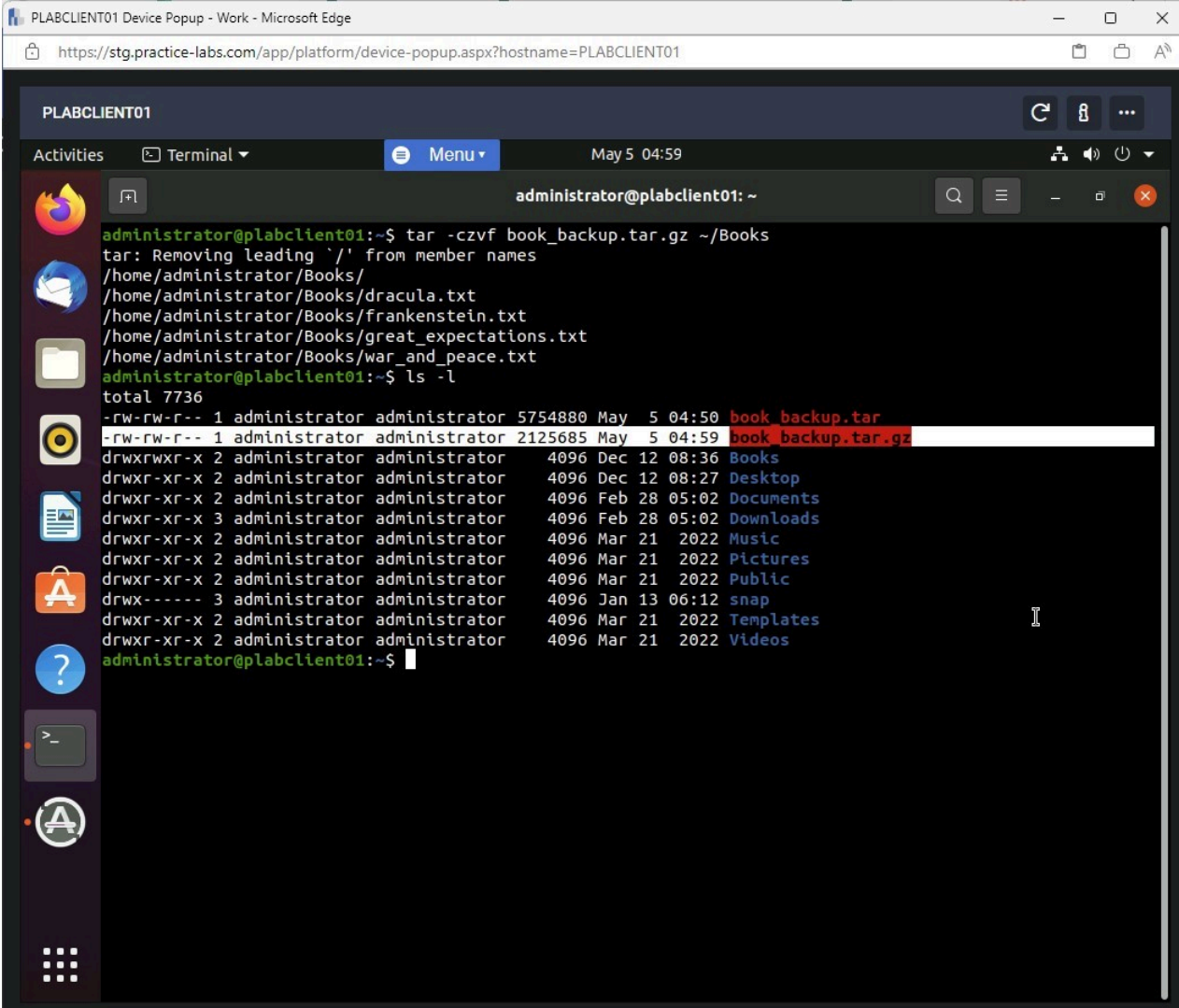
Step 8

In the **Terminal** window, to compress and archive the Books directory, type the following command and press **Enter**:

```
tar -czvf book_backup.tar.gz ~/Books
```

Next, type the following command and press **Enter**:

```
ls -l
```



The screenshot shows a terminal window titled 'PLABCLIENT01' with a dark background. The terminal displays the following commands and output:

```
administrator@plabclient01:~$ tar -czvf book_backup.tar.gz ~/Books
tar: Removing leading '/' from member names
/home/administrator/Books/
/home/administrator/Books/dracula.txt
/home/administrator/Books/frankenstein.txt
/home/administrator/Books/great_expectations.txt
/home/administrator/Books/war_and_peace.txt
administrator@plabclient01:~$ ls -l
total 7736
-rw-rw-r-- 1 administrator administrator 5754880 May  5 04:50 book_backup.tar
-rw-rw-r-- 1 administrator administrator 2125685 May  5 04:59 book_backup.tar.gz
drwxrwxr-x 2 administrator administrator 4096 Dec 12 08:36 Books
drwxr-xr-x 2 administrator administrator 4096 Dec 12 08:27 Desktop
drwxr-xr-x 2 administrator administrator 4096 Feb 28 05:02 Documents
drwxr-xr-x 3 administrator administrator 4096 Feb 28 05:02 Downloads
drwxr-xr-x 2 administrator administrator 4096 Mar 21 2022 Music
drwxr-xr-x 2 administrator administrator 4096 Mar 21 2022 Pictures
drwxr-xr-x 2 administrator administrator 4096 Mar 21 2022 Public
drwx----- 3 administrator administrator 4096 Jan 13 06:12 snap
drwxr-xr-x 2 administrator administrator 4096 Mar 21 2022 Templates
drwxr-xr-x 2 administrator administrator 4096 Mar 21 2022 Videos
administrator@plabclient01:~$
```

Figure 1.8 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered.

Note: Notice the file size for **book_backup.tar.gz**. The archive file is now compressed. The **z** parameter will create a new tar file using a gzip compression, and **v** is used to print information for the tar operation on the terminal window.

Step 9

In the **Terminal** window, type the following command and press **Enter**:

```
tar -xzvf book_backup.tar.gz
```

Next, type the following command and press **Enter**:

```
ls -l
```

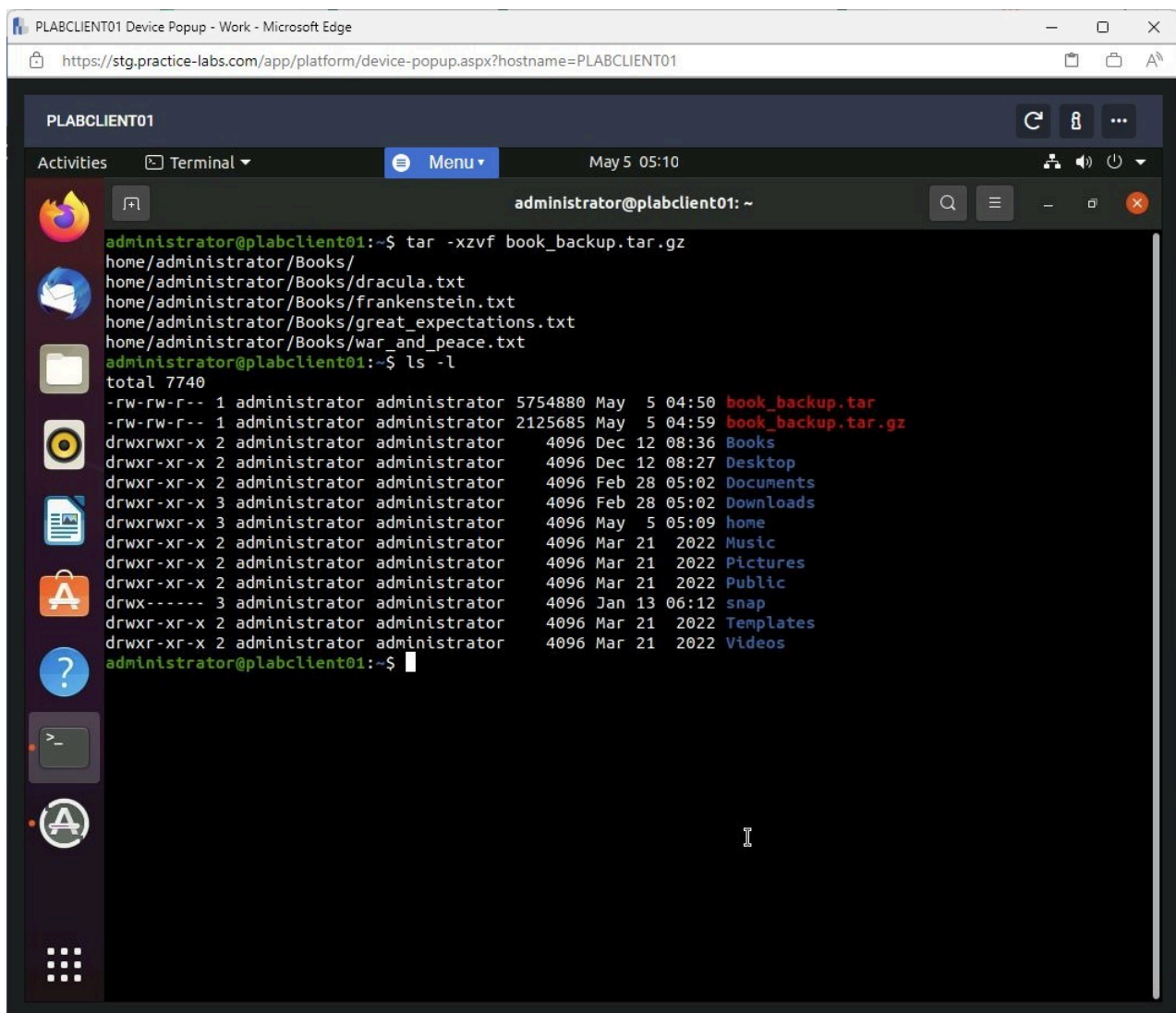


Figure 1.9 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered.

Note: The tar command used with the x option will extract the archive file. Notice the output of the tar command shows the location of files that have been restored.

Step 10

In the **Terminal** window, type the following command and press **Enter**:

clear

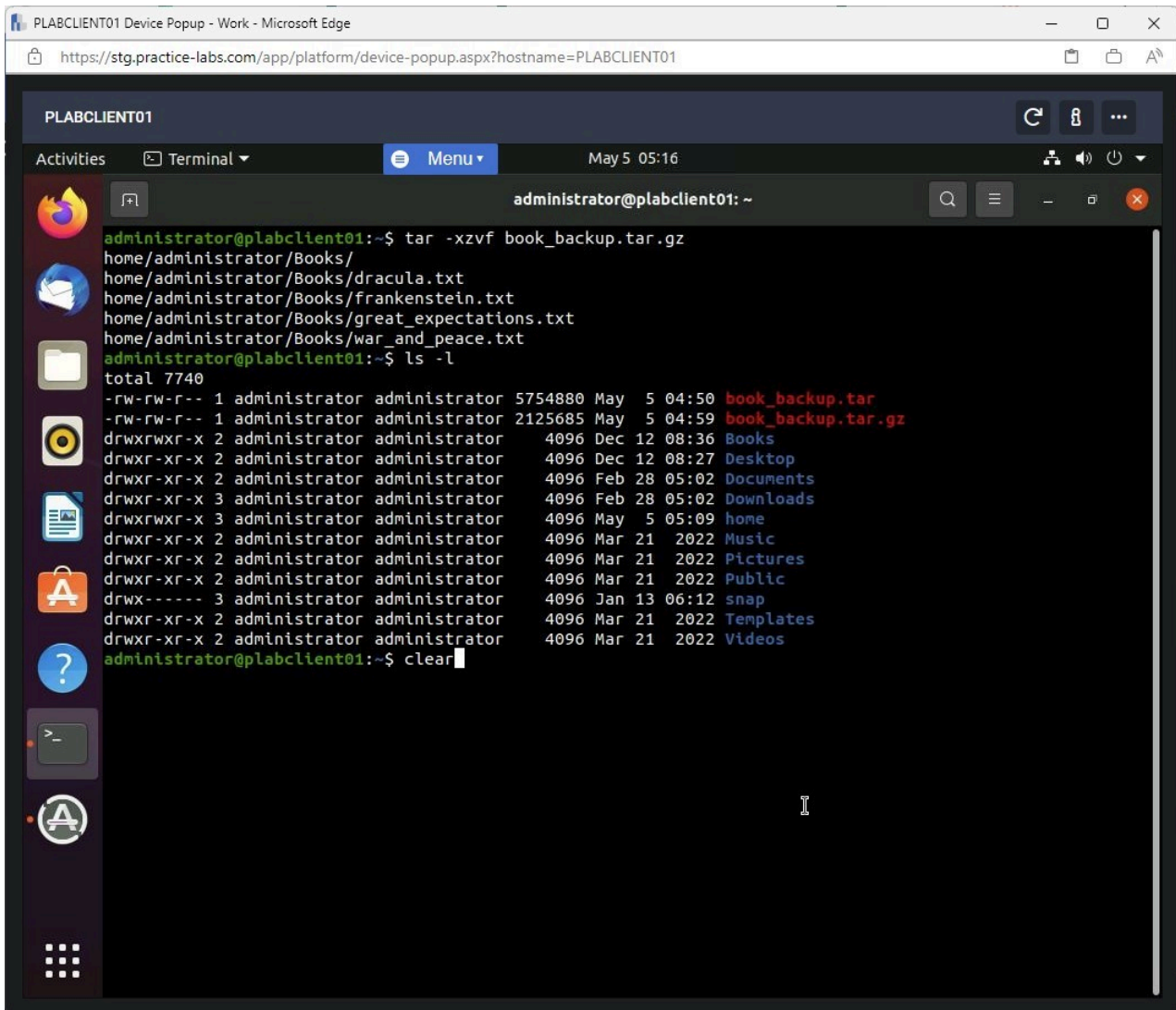


Figure 2.11 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered.

Note: The *clear* command will clear the Terminal window.

Keep the **Terminal** window open.

Exercise 2 - Use the Dd Command to Backup an Entire Disk

The main purpose of the `dd` command in Linux is to convert and copy files. In Linux, device drivers for disk drives, such as hard drives, are in the file system, just like other files. The `dd` command can be used to backup an entire disk because it can read/write from these files.

In this exercise, you will use the `dd` command to backup an entire disk.

Learning Outcomes

After completing this exercise, you should be able to:

- Backup an Entire Disk using the `Dd` Command

Your Devices

You will be using the following device in this lab. Please power this on now.

- **PLABCLIENT01** - Ubuntu 20.04 - Stand-Alone Ubuntu Workstation (GUI)

Task 1 - Backup an Entire Disk using the `Dd` Command

In this task, you will first create a clone of a disk. You will then use the `dd` command to backup the disk when a full disk image is required.

Step 1

Ensure you are connected to **PLABCLIENT01** and the **Terminal** window is open.

In the **Terminal** window, type the following command and press **Enter**:

```
lsblk
```

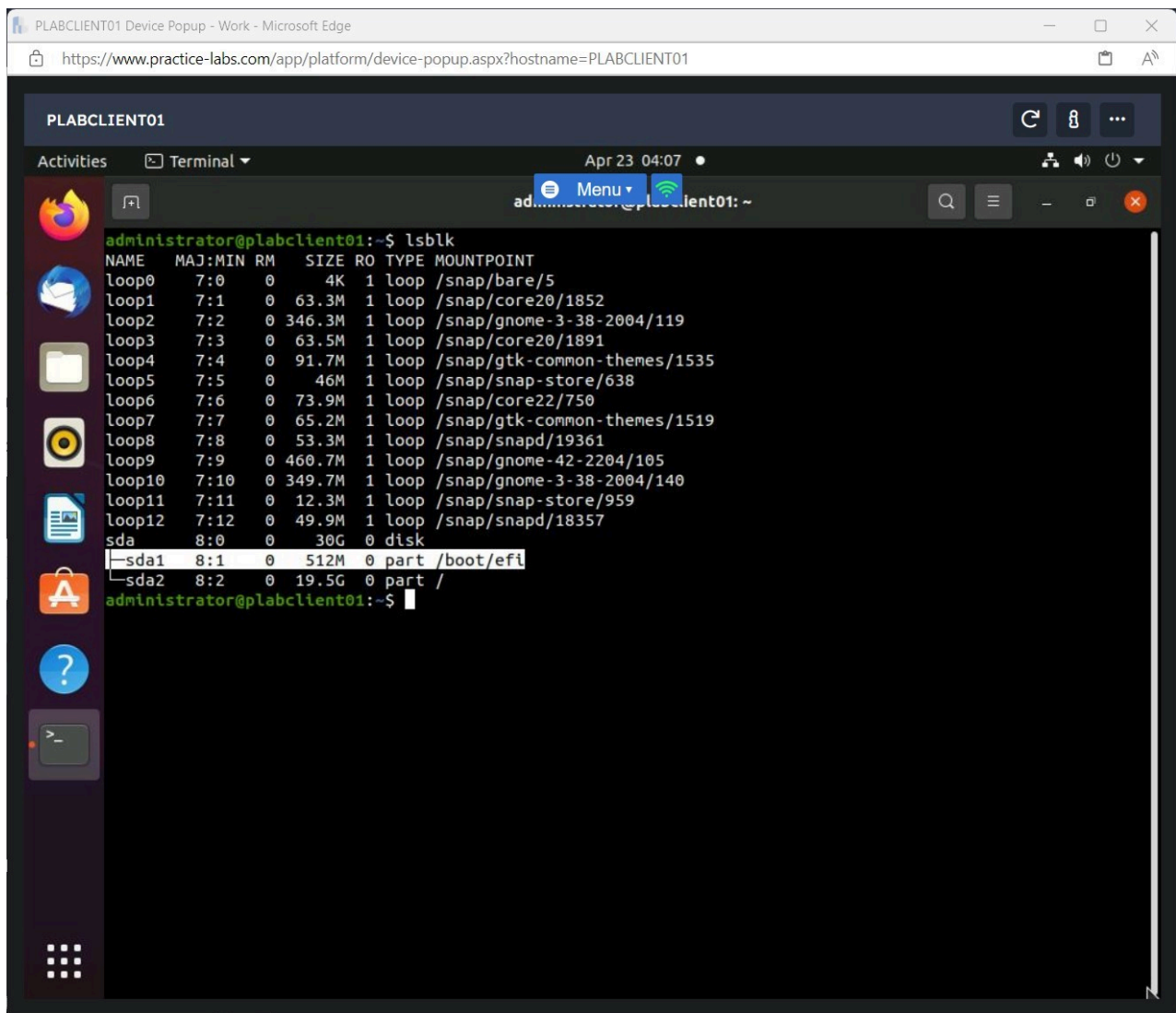


Figure 2.1 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The **lsblk** command will list information about all the block devices available. Notice the disk named **sda1**. You will backup this disk using the **dd** command. The **tar** command does not recognize volumes and cannot be used to backup disks.

Step 2

In the **Terminal** window, type the following command and press **Enter**:

```
sudo dd if=/dev/sda1 of=~ /cdrom.img bs=1M
```

Type the following **Password** and press **Enter**:

Passw0rd

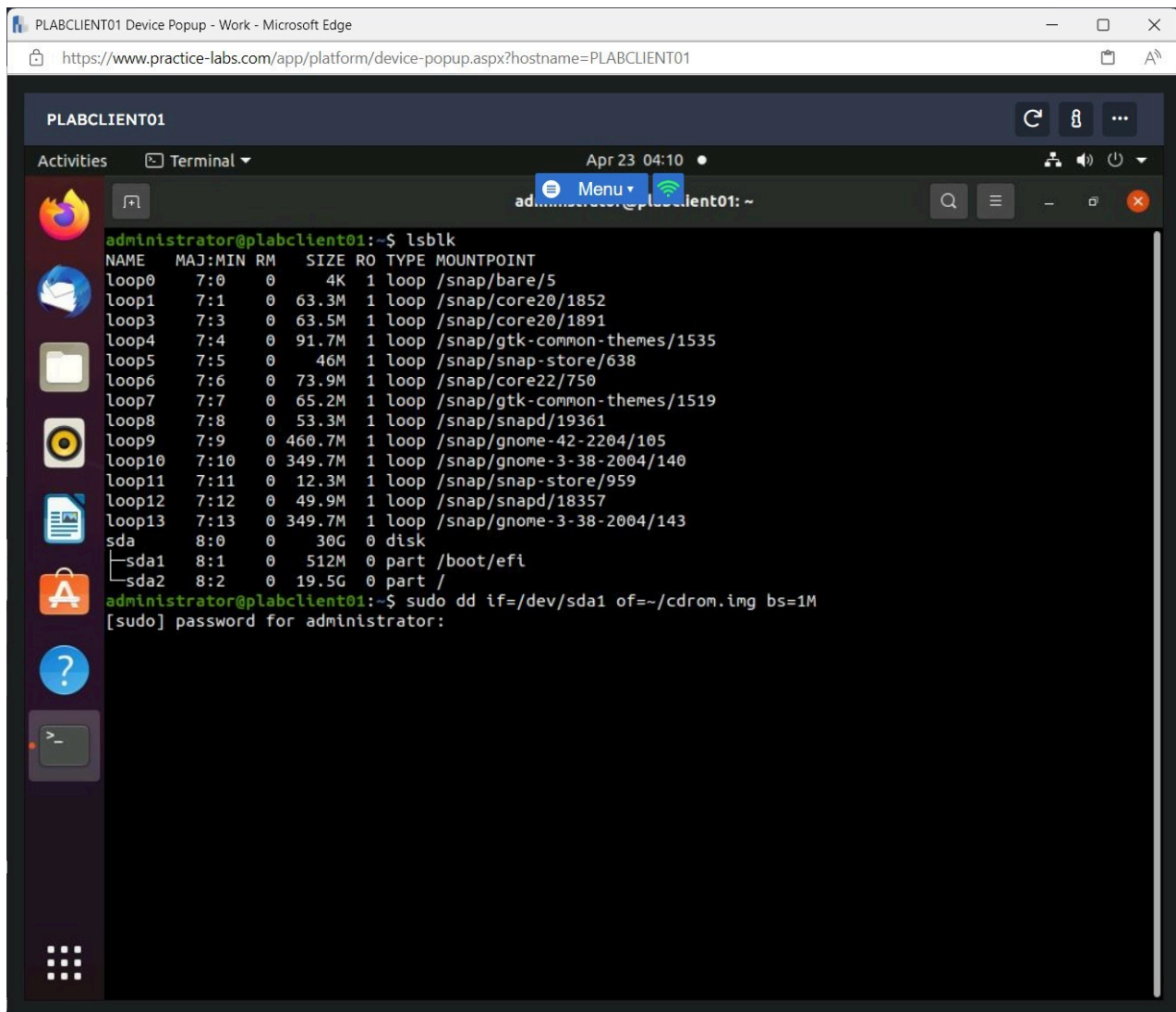


Figure 2.2 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The **dd** command has the following options:

- *if* = input file
- *of* = output file
- *bs* = block size - how much data it should read at a time

Restoring a backup is reversing the input file and output file: `sudo dd if=./cdrom.img of=/dev/sda`.

This step may take a few minutes to complete.

Step 3

In the **Terminal** window, type the following command and press **Enter**:

```
ls -l
```

```

PLABCLIENT01 Device Popup - Work - Microsoft Edge
https://www.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABCLIENT01

PLABCLIENT01
Activities Terminal Apr 23 04:10
administrator@plabclient01:~$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINT
loop0 7:0 0 4K 1 loop /snap/bare/5
loop1 7:1 0 63.3M 1 loop /snap/core20/1852
loop3 7:3 0 63.5M 1 loop /snap/core20/1891
loop4 7:4 0 91.7M 1 loop /snap/gtk-common-themes/1535
loop5 7:5 0 46M 1 loop /snap/snap-store/638
loop6 7:6 0 73.9M 1 loop /snap/core22/750
loop7 7:7 0 65.2M 1 loop /snap/gtk-common-themes/1519
loop8 7:8 0 53.3M 1 loop /snap/snapd/19361
loop9 7:9 0 460.7M 1 loop /snap/gnome-42-2204/105
loop10 7:10 0 349.7M 1 loop /snap/gnome-3-38-2004/140
loop11 7:11 0 12.3M 1 loop /snap/snap-store/959
loop12 7:12 0 49.9M 1 loop /snap/snapd/18357
loop13 7:13 0 349.7M 1 loop /snap/gnome-3-38-2004/143
sda 8:0 0 30G 0 disk
├─sda1 8:1 0 512M 0 part /boot/efi
└─sda2 8:2 0 19.5G 0 part /
administrator@plabclient01:~$ sudo dd if=/dev/sda1 of=~/cdrom.img bs=1M
[sudo] password for administrator:
512+0 records in
512+0 records out
536870912 bytes (537 MB, 512 MiB) copied, 1.08875 s, 493 MB/s
administrator@plabclient01:~$ ls -l
total 524332
drwxrwxr-x 2 administrator administrator 4096 Dec 12 2022 Books
-rw-r--r-- 1 root root 536870912 Apr 23 04:10 cdrom.img
drwxr-xr-x 2 administrator administrator 4096 Dec 12 2022 Desktop
drwxr-xr-x 2 administrator administrator 4096 Feb 28 2023 Documents
drwxr-xr-x 3 administrator administrator 4096 Feb 28 2023 Downloads
drwxr-xr-x 2 administrator administrator 4096 Mar 21 2022 Music
drwxr-xr-x 2 administrator administrator 4096 Mar 21 2022 Pictures
drwxr-xr-x 2 administrator administrator 4096 Mar 21 2022 Public
drwx----- 3 administrator administrator 4096 Jan 13 2023 snap
drwxr-xr-x 2 administrator administrator 4096 Mar 21 2022 Templates
drwxr-xr-x 2 administrator administrator 4096 Mar 21 2022 Videos
administrator@plabclient01:~$

```

Figure 2.3 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: Notice the size of the output file. Ignore the error message that appears. To reduce the size, the file can be compressed, but it is a separate command using `dd`. The output file must be piped (`|`) into the `gzip` command.

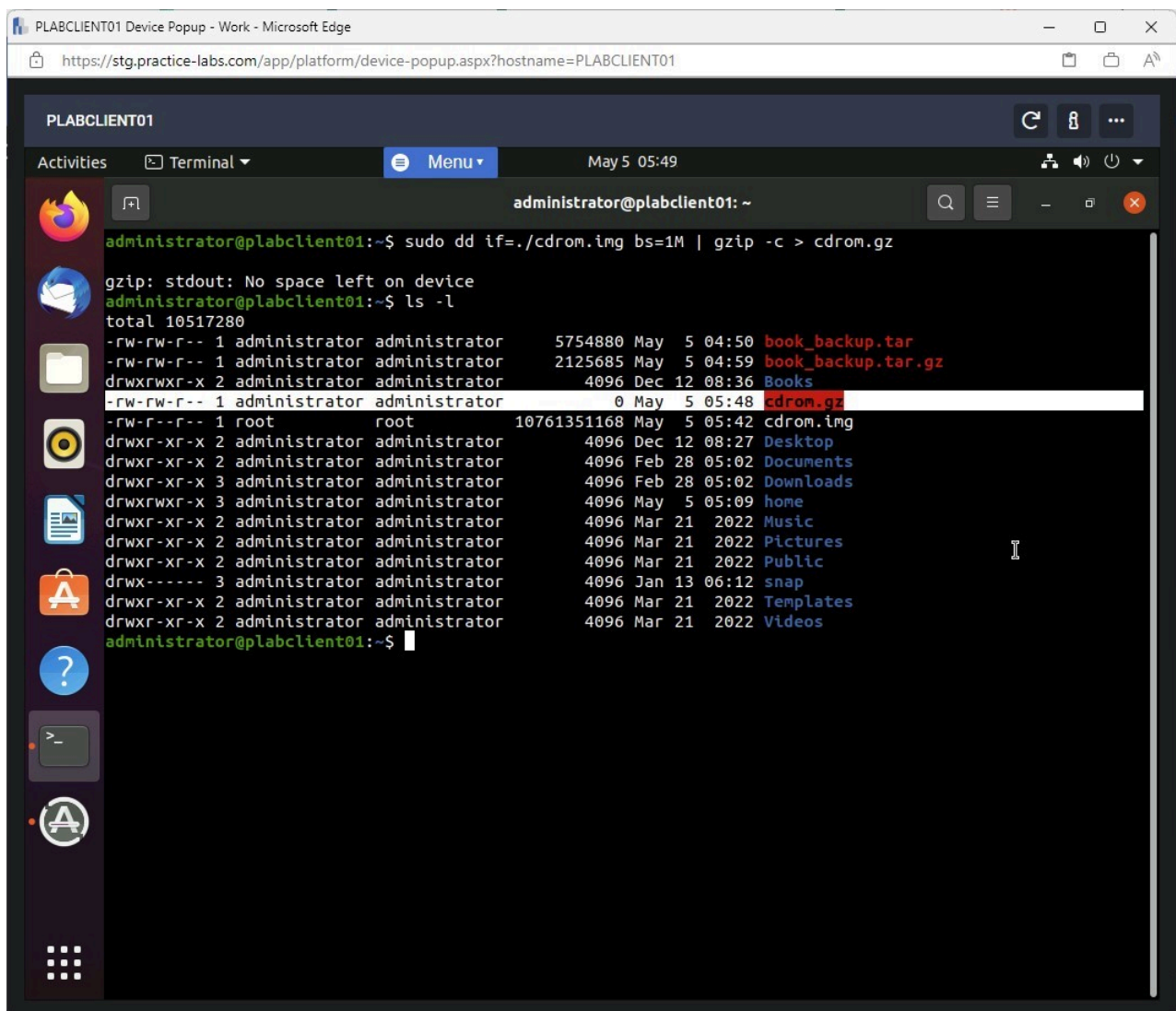
Step 4

In the **Terminal** window, type the following command and press **Enter**:

```
sudo dd if=./cdrom.img bs=1M | gzip -c > cdrom.gz
```

Next, type the following command and press **Enter**:

```
ls -l
```



The screenshot shows a terminal window titled "PLABCLIENT01" with a dark background. The prompt is "administrator@plabclient01: ~". The command "sudo dd if=./cdrom.img bs=1M | gzip -c > cdrom.gz" has been entered and executed. The output shows "gzip: stdout: No space left on device". Below this, the command "ls -l" has been entered, and the output is a long listing of files and directories. The file "cdrom.gz" is highlighted in red in the original image. The listing shows the following files and directories:

Permissions	Size	Owner	Group	Size	Month	Day	Time	File/Directory
-rw-rw-r--	1	administrator	administrator	5754880	May	5	04:50	book_backup.tar
-rw-rw-r--	1	administrator	administrator	2125685	May	5	04:59	book_backup.tar.gz
drwxrwxr-x	2	administrator	administrator	4096	Dec	12	08:36	Books
-rw-rw-r--	1	administrator	administrator	0	May	5	05:48	cdrom.gz
-rw-r--r--	1	root	root	10761351168	May	5	05:42	cdrom.img
drwxr-xr-x	2	administrator	administrator	4096	Dec	12	08:27	Desktop
drwxr-xr-x	2	administrator	administrator	4096	Feb	28	05:02	Documents
drwxr-xr-x	3	administrator	administrator	4096	Feb	28	05:02	Downloads
drwxrwxr-x	3	administrator	administrator	4096	May	5	05:09	home
drwxr-xr-x	2	administrator	administrator	4096	Mar	21	2022	Music
drwxr-xr-x	2	administrator	administrator	4096	Mar	21	2022	Pictures
drwxr-xr-x	2	administrator	administrator	4096	Mar	21	2022	Public
drwx-----	3	administrator	administrator	4096	Jan	13	06:12	snap
drwxr-xr-x	2	administrator	administrator	4096	Mar	21	2022	Templates
drwxr-xr-x	2	administrator	administrator	4096	Mar	21	2022	Videos

Figure 2.4 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: Notice the size of the compressed `cdrom.gz` file. Your result maybe different to the screenshot due to time and updates implemented, however the key thing to note is the file size has been compressed. You may have also noted some errors were reported. Below are some options to address those errors:

- **`conv=noerror`** can be used to ignore errors

- **conv=sync** will ensure a perfect bit by bit copy but slows down the process because it checks all writes before writing the file.

To unzip the compressed file, the **gunzip** command is used as follows to run a similar reversal:

```
gunzip -c ~/if=./cdrom.gz of=/dev/sda bs=1M
```

Close the **Terminal** window.

Keep all devices that you have powered on in their current state and proceed to the review section.

Review

Well done, you have completed the **Linux Backup and File Compression Concepts** Practice Lab.

Summary

You completed the following exercises:

- Exercise 1 - Use Tar to Backup and Compress Files
- Exercise 2 - Use the Dd Command to Backup an Entire Disk

You should now be able to:

- Use the Tar Command to Archive and Compress Files
- Backup an Entire Disk using the Dd Command

Feedback

Shutdown all virtual machines used in this lab. Alternatively, you can log out of the lab platform.